

Evidence Behind the Schmitt-Thompson Adult Telehealth Triage Guidelines



Overview

The 2025 Schmitt-Thompson Clinical Content (STCC) contains 446 Adult After Hours telehealth triage and advice guidelines.

The following evidence-based resources are used during the development of new guidelines and the annual updating of existing guidelines.

Recent National Guidelines

New and updated guidelines from the AAP, AAFP, ACEP, ACOG, AHA, CDC, FDA, ISDA, USPSTF, and other national organizations are incorporated into the STCC guidelines in a timely manner. Here are several examples.

- In **2025** the Centers for Disease Control and Prevention updated their recommendations for *Antiretroviral Postexposure Prophylaxis After Sexual, Injection Drug Use, or Other Nonoccupational Exposure to HIV*.¹
- In **2023**, the United States Federal Drug Administration (FDA) approved the nasal form of Narcan to be sold over-the-counter without a prescription. First Aid Care Advice in multiple adult and pediatric telehealth triage guidelines, including Opioid Use and Problems, provides step-by-step instructions on how to use Narcan.²
- The Centers for Disease Control and Prevention (CDC) is the primary source for diagnosis and treatment recommendations for Sexually Transmitted Infections. Multiple adult and pediatric telehealth triage guidelines, such as *STI Questions* and *Vaginal Discharge*, reflect the CDC's **2021** guidance.³

¹ Tanner MR, O'Shea JG, Byrd KM, et.al. Antiretroviral Postexposure Prophylaxis After Sexual, Injection Drug Use, or Other Nonoccupational Exposure to HIV - CDC Recommendations, United States, 2025. MMWR Recomm Rep. 2025 May 8;74(1):1-56. PMID: 40331832. Available at <https://pmc.ncbi.nlm.nih.gov/articles/PMC12064164/>.

² Available at [FDA Approves First Over-the-Counter Naloxone Nasal Spray | FDA](https://www.fda.gov/news-events/press-announcements/fda-approves-first-over-the-counter-naloxone-nasal-spray).

³ Workowski KA, Bachmann LH, Chan PA, Johnston CM, Muzny CA, Park I, Reno H, Zenilman JM, Bolan GA. Sexually Transmitted Infections Treatment Guidelines, 2021. MMWR Recomm Rep. 2021 Jul 23;70(4):1-187. Available at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8344968/>.

- The National Asthma Education and Prevention Program (NAEPP) is coordinated and supported by the National Heart, Lung, and Blood Institute (NHLBI) of the National Institutes of Health. NAEPP's goal is to improve patient care and provide informed decision making about asthma management in the clinical setting. The adult and pediatric *Asthma Attack* telehealth triage guideline incorporate the most recent **2020** NAEPP recommendations. For example, triage guideline Care Advice provides guidance that during an asthma attack a person should use their short-acting quick-relief inhaler (4 puffs, such as albuterol) or nebulizer up to 3 times, every 20 minutes as needed.⁴
- The Wilderness Medicine Society (WMS) is a key source for clinical information about the prevention and treatment of altitude illness. The adult *Altitude Sickness* telehealth triage guidelines incorporates the latest **2019** WMS definitions and recommendations.⁵
- The American College of Obstetricians and Gynecologists (ACOG) provides education and publishes clinical practice guidelines on pregnancy and postpartum care. The adult *Pregnancy - Morning Sickness (Nausea and Vomiting of Pregnancy)* telehealth triage guideline integrates information from ACOG's **2018** Practice Bulletin.⁶
- Since 1977 the American College of Cardiology (ACC) and the American Heart Association (AHA) have jointly published a series of reports on the evaluation and management of high blood pressure in adults. The adult *Blood Pressure – High* telehealth triage guideline reflects the recommendations and blood pressure thresholds from the most recent **2017** report.⁷
- The guidelines support antibiotic stewardship. STCC emphasizes the importance of not prescribing antibiotics for viral respiratory infections. The guidelines also support other objectives of the **2012** Choosing Wisely campaign, an initiative of the American

⁴ Expert Panel Working Group of the National Heart, Lung, and Blood Institute (NHLBI) administered and coordinated National Asthma Education and Prevention Program Coordinating Committee (NAEPCC); Cloutier MM, Baptist AP, et.al. 2020 Focused Updates to the Asthma Management Guidelines: A Report from the National Asthma Education and Prevention Program Coordinating Committee Expert Panel Working Group. J Allergy Clin Immunol. 2020 Dec;146(6):1217-1270. Available at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7924476/>.

⁵ Luks AM, Auerbach PS, Freer L, et.al. Wilderness Medical Society Clinical Practice Guidelines for the Prevention and Treatment of Acute Altitude Illness: 2019 Update. Wilderness Environ Med. 2019 Dec;30(4S):S3-S18.

⁶ Committee on Practice Bulletins-Obstetrics. ACOG Practice Bulletin No. 189: Nausea And Vomiting Of Pregnancy. Obstet Gynecol. 2018 Jan;131(1):e15-e30.

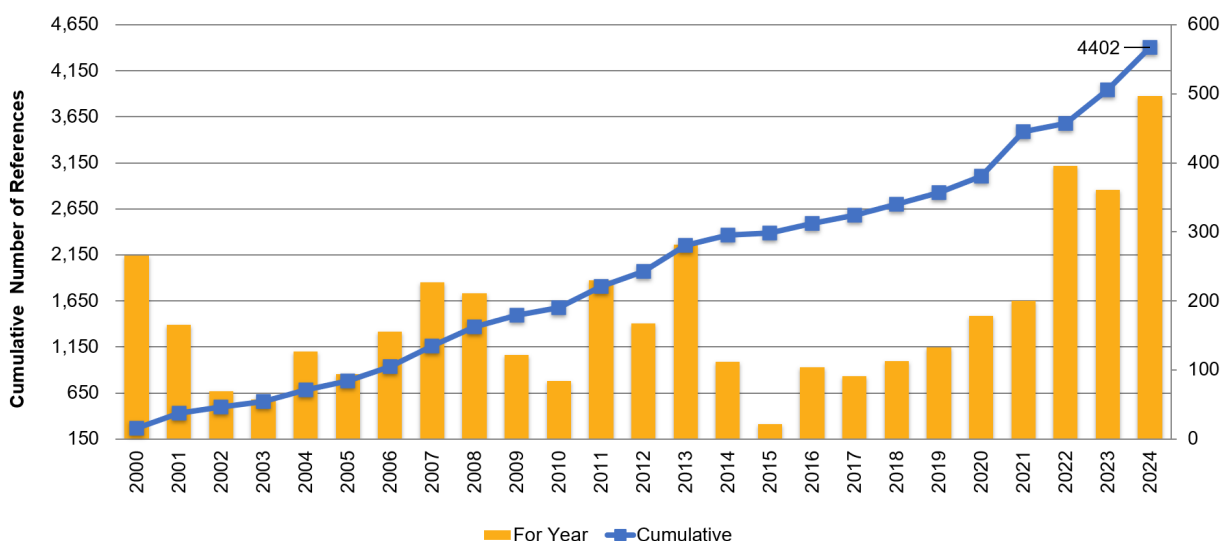
⁷ Whelton PK, Carey RM, Aronow WS, et.al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Hypertension. 2018 Jun;71(6):e13-e115. Available at <https://www.ahajournals.org/doi/pdf/10.1161/HYP.000000000000065>.

Board of Internal Medicine. We have also developed a number of disease-specific antibiotic follow-up guidelines such as *Cellulitis on Antibiotic Follow-up Call*.^{8 9}

Recent Medical Literature

Medical journals are reviewed on a quarterly basis by a work group of our physician and nurse members of the Clinical Advisory Committee. Members look for new research relevant to telehealth triage and advice.

Articles that expand and support the evidence base are logged and added to the clinical content database. More than a hundred new references are added to the guidelines each year. STCC includes all of these research citations in the database that we distribute to our clinical and software partners.



Clinical research findings and recommendations that improve the guideline content are incorporated into the triage decision-logic, Care Advice, and Background Area (health information). For example, the triage decision-logic in the *Head Trauma* guideline reflects

⁸ Harris AM, Hicks LA, Qaseem A; High Value Care Task Force of the American College of Physicians and for the Centers for Disease Control and Prevention. Appropriate Antibiotic Use for Acute Respiratory Tract Infection in Adults: Advice for High-Value Care From the American College of Physicians and the Centers for Disease Control and Prevention. *Ann Intern Med*. 2016 Mar 15;164(6):425-34. Available at https://www.acpjournals.org/doi/full/10.7326/M15-1840?rfr_dat=cr_pub++0pubmed&url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Aacrossref.org.

⁹ Choosing Wisely. An Initiative of the ABIM Foundation. Available at <https://choosingwisely.org/>.

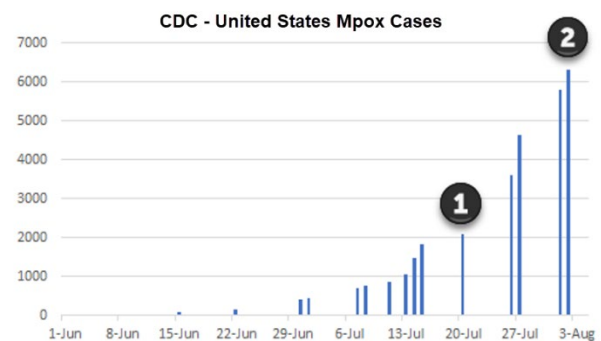
large prospective studies that established rules for predicting serious head injuries (Canadian CT Head Rule, New Orleans Criteria, Nexus Head CT).^{10 11 12 13}

Another example is the use of consensus recommendations such as sick day medication guidance for people with diabetes (PAUSE) in the *Diabetes – High Blood Sugar* and *Diabetes – Low Blood Sugar* guidelines.¹⁴

Responding to Infection Outbreaks and Public Health Emergencies

STCC is proactive and timely in responding to infectious disease outbreaks and other public health emergencies. Here are some examples.

- In July of **2022**, data from the Centers for Disease Control and Prevention (CDC) showed an upward trend in the Mpox case count in the United States, Canada, and across the globe. We immediately developed two new Adult Telehealth Triage Guidelines (Protocols) to provide decision-support to telehealth triagers and care advice for the patient populations they serve. We updated and distributed these guidelines again 2 weeks later when CDC guidance changed.



¹⁰ Stiell IG, Wells GA, Vandemheen K, et.al. The Canadian CT Head Rule for patients with minor head injury. *Lancet*. 2001 May 5;357(9266):1391-6.

¹¹ Haydel MJ, Preston CA, Mills TJ, et al. Indications for computed tomography in patients with minor head injury. *N Engl J Med*. 2000;343:100-105.

¹² Mower WR, Gupta M, Rodriguez R, et al. Validation of the sensitivity of the National Emergency X- Radiography Utilization Study (NEXUS) head computed tomographic (CT) decision instrument for selective imaging of blunt head injury patients: an observational study. *PLoS Med*. 2017;14:e1002313.

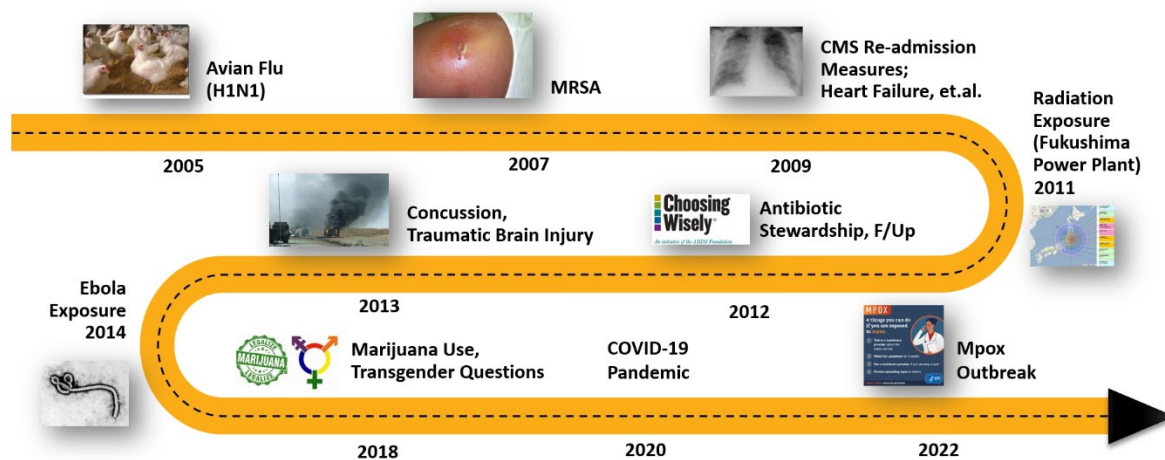
¹³ American College of Emergency Physicians Clinical Policies Subcommittee (Writing Committee) on Mild Traumatic Brain Injury; Valente JH, Anderson JD, Paolo WF, et.al. Clinical Policy: Critical Issues in the Management of Adult Patients Presenting to the Emergency Department With Mild Traumatic Brain Injury: Approved by ACEP Board of Directors, February 1, 2023 Clinical Policy Endorsed by the Emergency Nurses Association (April 5, 2023). *Ann Emerg Med*. 2023 May;81(5): e63-e105.

¹⁴ Watson KE, Dhaliwal K, Robertshaw S, Verdin N, Benterud E, Lamont N, Drall KM, McBrien K, Donald M, Tsuyuki RT, Campbell DJT, Pannu N, James MT; PAUSE (Preventing Medication Complications During Acute Illness Through Symptom Evaluation and Sick Day Guidance) Medication Safety Advisory Panel. Consensus Recommendations for Sick Day Medication Guidance for People With Diabetes, Kidney, or Cardiovascular Disease: A Modified Delphi Process. *Am J Kidney Dis*. 2023 May;81(5):564-574.

- In **2020**, within two weeks of the start of the COVID-19 pandemic, STCC published a COVID-19 Exposure guideline. This was soon followed by three other COVID-19 guidelines: Diagnosis Suspected, Persisting Symptoms, and Vaccine Questions and Reactions guidelines. All were reviewed by the CDC and infectious disease specialists. Regular updates were released as national policies changed and new information about the virus became available. During the COVID-19 pandemic, STCC published and released more than a dozen updates on each guideline.

STCC remains vigilant for any crisis guidelines that might be needed, writes them quickly, and distributes them to our software and clinical partners.

Other examples over the past two decades include Avian Flu (H1N1), CMS Hospital Readmission Measures (e.g., Heart Failure), Ebola, E-Cigarettes and Vaping Symptoms and Questions, MRSA, Opioid Use and Problems, Radiation Exposure, and many more.



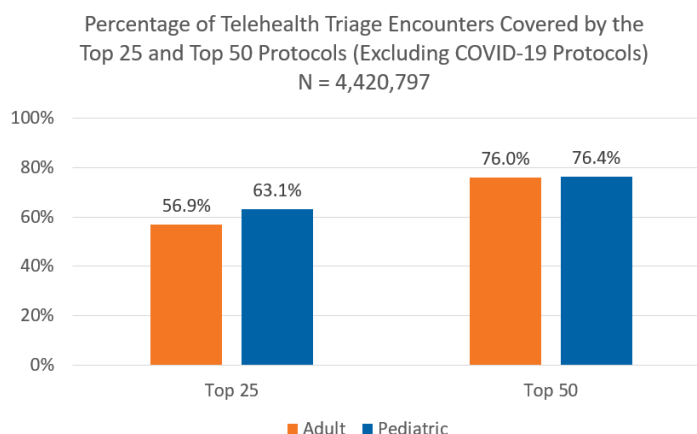
Research on STCC Guidelines

A number of studies regarding the use of the STCC telehealth triage guidelines have been published in peer-reviewed journals. Examples include:

- Study of 900,196 telephone triage encounters demonstrated that triaged patients were sicker or more injured than people who presented directly to the emergency department without triage.¹⁵
- During the two study periods that were a total of xxx telehealth triage calls to a medical call center serving the a province in Canada. According to study criteria, most referrals to the ED were appropriate (84%, 87%). The COVID-19 pandemic did not significantly change the number of Health Link calls leading to ED referrals or the appropriateness of those referrals.¹⁶
- Patient safety study that documented a 90% appropriateness of pediatric ED referrals.¹⁷
- Cost-effectiveness study of pediatric call centers showed that four medical care dollars were saved for every one dollar spent.¹⁸

Much of STCC research is proprietary and unpublished. Research has included large data analyses of age, physiologic gender, reason for visit, pain score, and temperature on need for hospital admission.¹⁹

We have also partnered with our clinical and software partners to study triage guideline utilization, triage disposition



¹⁵ Schmaus A, Cooper IR, Whitten T, Bakal J, Watt D, Huang J, Lang E. Impact of Health Link utilization on emergency department visits. CJEM. 2023 Apr 23. Epub ahead of print. [PubMed Abstract](#). [Full Text in Canadian Journal of Emergency Medicine](#).

¹⁶ Cooper IR, Schmaus A, Whitten TA, Bakal JA, Kurji F, Watt D, Lang E. Descriptive analysis and evaluation of Health Link referrals to the emergency department before and during the COVID-19 pandemic. Intern Emerg Med. 2024 Feb 22. [PubMed Abstract](#).

¹⁷ Kempe A, Dempsey C, Whitefield J, Bothner J, MacKenzie T, Poole S. Appropriateness of urgent referrals by nurses at a hospital-based pediatric call center. Arch Pediatr Adolesc Med. 2000;154(4):355-360. [PubMed Abstract](#).

¹⁸ Bunik M, Glazner JE, Chandramouli V, Emsermann CB, Hegarty T, Kempe A. Pediatric telephone call centers: how do they affect health care use and costs? Pediatrics. 2007;119(2):e305-e313. [PubMed Abstract](#).

¹⁹ Thompson DA. Temperature Thresholds, Age Group, and the Effect on Emergency Department Admission Rate. Analysis of 34,936 Emergency Department patients. NHAMCS 2010 Data Set. Centers for Disease Control and Prevention (www.cdc.gov). Unpublished data. 2014.

rates, and pre-call intent vs nurse triage recommended level of care.^{20 21 22 23}

A full list of published and unpublished studies is available on request.

Review by Clinical Advisory Committee and Subject Matter Experts

All **new guidelines** are developed using an 11-step process following our standardized clinical governance procedures. New guidelines are rigorously reviewed by multiple nurse and physician members of our *Review Panel*, which includes: our Editorial Team, our Clinical Advisory Committee, Specialty Work Groups, and other clinical consultants and subject matter experts.

As part of our ongoing review process **existing guidelines** are reviewed, updated, and then annually sent out to our clinical and software partners. This update process ensures that our guidelines are current, evidence-based, and compliant with the latest recommendations from national organizations.



²⁰ Marks GA, Thompson DA. Top 25 Adult and Pediatric Office-Hours Telehealth Triage Guidelines Used During Calendar Years 2020 - 2022. March 31, 2023. [Abstract and Chart](#).

²¹ Thompson DA. Medical Call Center Benchmarking Report: 2004 to 2016. An analysis of 6.5 million triage calls from 10 medical call centers. Limited distribution to participating call centers. 2017.

²² Brown K, Tabone G, Thompson DA. The impact of telephone triage on healthcare costs – an analysis of caller intent and outcomes. 2018. [Abstract and Chart](#).

²³ O'Bryan L, Barzin A, Thompson DA. Comparison of Pre-Call Patient Intent to Telehealth Triage Nurse Recommended Disposition. May 17, 2023. [Abstract and Charts](#).

Each year we provide “redline” comparisons (to the prior year) of all guidelines to clearly show our clinical partners exactly what minor and major changes have been made from year-to-year. We also write a detailed annual letter to triage nurses, detailing a summary of changes, and serving as a self-study guide for the telehealth triage clinical content.

A full list of members of our *Review Panel* and a document describing our *Content Development and Update* process are available on request.



Respected and Recommended

The STCC telehealth triage guidelines have been used and clinically validated through usage by over 400 medical call centers and thousands of doctor’s offices. For over 25 years the STCC guidelines have been updated, improved, and expanded based on clinical use by healthcare organizations across North America.

The STCC telehealth triage guidelines are used by the top hospitals in the United States.



Summary

Schmitt-Thompson Clinical Content (STCC) is a comprehensive resource featuring nearly 800 evidence-based telehealth triage guidelines for both adults and children. STCC swiftly integrates new data and national clinical guidelines to address emerging health challenges such as Mpox, COVID-19, and Ebola. The guidelines are developed through rigorous clinical governance processes involving expert clinicians and are continuously updated based on recent medical literature and research findings. These guidelines have demonstrated their effectiveness in improving patient safety and cost-efficiency in several studies. Trusted by over 400 medical call centers and top hospitals in the United States, STCC's guidelines represent a reliable, evidence-based, tested tool in the rapidly evolving field of telehealth.